



WORK COMPLETED BY

SOHAM PHULARE

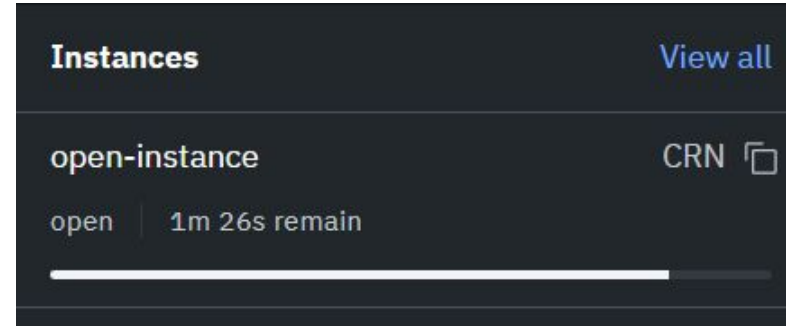
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IBM QPU TIME ESTIMATION

Objective: To determine a formula to estimate time taken by the QPU to simulate a molecule.

Need for time estimation

- On IBM Quantum Platform the free plan only provides 10 minutes of Quantum Runtime.
- These 10 minutes very quickly becomes the limiting factor when working with larger and more complex molecules
- In order to solve this issue we can request IBM to provide some Quantum Runtime for research purposes.
- For this reason we are to estimate the QPU time by creating a formula.



Runtime Usage on IBM Quantum Platform

Variables Driven by Hamiltonian & Active Space

The Hamiltonian and the chosen active space define which operators must be measured and how many distinct runs of the ansatz are needed.

- **N_c**: number of mutually commuting groups of Pauli strings, determined by the mapped electronic Hamiltonian.
- Larger active spaces increase term count and qubit count, which typically inflates N_c.
- Different mappers decompose the same Hamiltonian into different Pauli strings, changing N_c.

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Group 1 Pauli Strings:['ZZ', 'XX', 'II']  
Group 2 Pauli Strings:['XZ', 'ZX']  
Group 3 Pauli Strings:['IZ', 'ZI']  
Group 4 Pauli Strings:['IX', 'XI']
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Groups of Pauli Strings for LiH
in (2e,2o) Active Space

Variables Driven by Ansatz and Transpiler

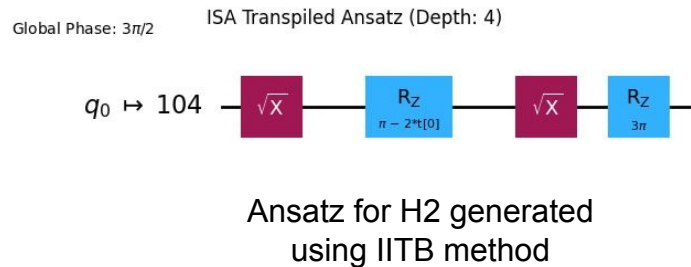
Once the Hamiltonian is fixed, the ansatz and transpiler decide how deep each run is and how many times it must be repeated.

D : critical-path depth of the transpiled circuit, including the ansatz and basis-rotation gates.

- Complex ansätze over many qubits give large base depths; the transpiler adds SWAPs, increasing D further.

N_{shots} : number of repetitions per measurement basis, set at the Estimator/Runtime level.

- Shallow, hardware-efficient ansätze may tolerate lower shots; deeper, noisier circuits often need more shots to maintain statistical quality.



Variables Driven by Backend (QPU)

Once the algorithmic structure is fixed, the backend hardware determines the physical time scales.

Hardware constants:

- T_{gate} : microwave pulse length for single-/two-qubit gates.
- T_{readout} : time to resolve the qubit state in the readout resonator.
- T_{latency} : overhead from control electronics and FPGA coordination.

Rep-delay and initialization:

- T_{delay} : A brief pause between shots to let the hardware clear out old signals.
- T_{init} : The time it takes to actively force the qubits back to zero. This depends entirely on the slowest qubit you are using.

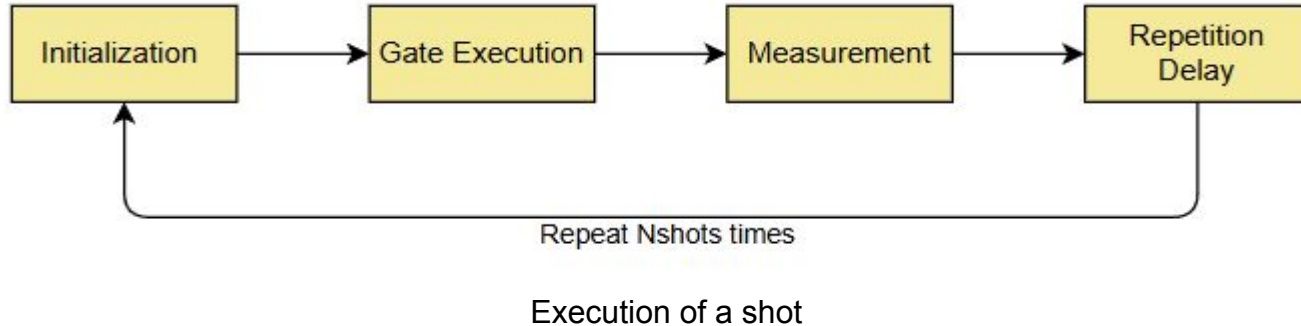
T_{gate}	68.43 ns
T_{readout}	2.28 μs
T_{delay}	250 μs
T_{init}	1500 μs

QPU Time Formula

By combining all of the above factors, we get the following formula:

$$T_{total} = N_c \times N_{shots} \times [(D \times T_{gate}) + T_{init} + T_{delay} + T_{readout}]$$

We can use the formula to verify time by comparing it with time taken by real runs



Results

Molecule	No of Shots	QPU Time (actual)	QPU Time (calculated)
H2O	4000	141 s	137 s
H2O	2000	74 s	69 s
H2O	1000	36 s	34 s

Results for formula achieved by simulating H2O molecule using varying number of shots

(All numbers have been rounded to the nearest whole number.)

As we can see from the above results the formula currently under-shoots the actual value for the QPU time.

Disclaimer : The above QPU runs have been done using only 1 iteration

Results (Continued)

Molecule	No of Shots	QPU Time (actual)	QPU Time (calculated)
H2	4000	3 s	14 s
H2	2000	2 s	7 s
H2	1000	2 s	4 s

Results for formula achieved by simulating H2 molecule using varying number of shots

(All numbers have been rounded to the nearest whole number.)

As we can see from the above results the formula currently significantly over-shoots the actual value for the QPU time.

Disclaimer : The above QPU runs have been done using only 1 iteration

THANK YOU